Airbnb-Scale Vacation Rental Marketplace Architecture

Production-Grade Distributed System Design: 100M+ Listings | 150K QPS | Zero Overbooking

1. Global Anycast Edge

Cloudflare & Akamai WAF

- 300+ Edge PoP Anycast Ingress
- DDoS L3/L4/L7 Mitigation
- TLS 1.3 & HTTP/3 Termination
- Edge Static Asset Caching

2. Edge Web / Next.js SSR

Vercel / Cloudflare Workers

- Dynamic ISR Page Rendering
- Device Viewport Responsive Image CDN
- Edge Auth & Token Verification
- Localized Geo-IP Routing

3. API Gateway & Envoy Mesh

Envoy Gateway / Kubernetes Ingress

- mTLS Mutual Authentication
- Dynamic Rate Limiting (Token Bucket)
- GraphQL & gRPC Federation
- Distributed Tracing (OpenTelemetry)

Listing Service

Metadata & Bento Grid

- Photos & Rooms
- Amenities Engine
- Host Policies
- House Rules

Availability Engine

Double-Booking Shield

- Calendar Bitmasks
- Slot Reservation
- Lock Coordination
- Sync Events

Search & Discovery

Geo-Spatial Engine

- Uber H3 Hex Index
- Facet Multi-Filtering
- Price Range Filters
- Sub-45ms P99

Pricing & Quotes

Dynamic Billing Engine

- Base Night Rates
- Cleaning & Service Fees
- Local Occupancy Tax
- Discount Coupons

Reviews & Trust

Reputation Engine

- 6-Category Ratings
- Verified Stays Only
- Superhost Badging
- Sentiment Scoring

Apache Kafka Real-Time Event Backbone & Debezium CDC

High-Throughput Partitioned Event Stream (Zero Data Loss)

- Event Topics: booking-initiated | booking-confirmed | calendar-locked | price-updated | inventory-sync
- Debezium CDC: Streams PostgreSQL WAL directly to Kafka for automatic cache & search sync without dual writes

Sharded PostgreSQL (Citus)

ACID Relational Core

- Sharded by Geography & Listing ID
- Row-level locking for checkout commits
- Multi-AZ synchronous replication
- Zero data loss transactional guarantees

Redis Enterprise Cluster

In-Memory Distributed Locks

- Redlock algorithm for 10s checkout locks
- Active calendar bitmasks in memory
- Sub-millisecond read latency
- Listing metadata L2 cache

OpenSearch Geo Cluster

Geospatial Discovery

- Uber H3 spatial index for map clusters
- Fast multi-faceted listing search
- Real-time inventory filter sync
- Instant full-text query matching